Theory of Equations

The Polynomial

If n is a positive integer, and $c_0, c_1, c_2, c_3, ..., c_n$ are complex constants belonging to $\mathbb{C}$ then

$$\boxed{f(x) \equiv c_0x^n + c_1x^{n-1} + c_2x^{n-2} + ... c_{n-1}x + c_n = \sum_{k=0}^{n} c_kx^{n-k}}$$

The Remainder Theorem

Theorem:

> $States\ that\ if\ the\ polynomial\ f(x)\ is\ divided\ by\ the\ linear\ factor\ (x-c)\ until\ a\ remainder\ independent\ of\ x\ is\ obtained\ then\ the\ remainder\ is\ f(c)$

Proof:

From the polynomial division alogorithm we obtain $f(x) \equiv q(x)g(x) + r(x)$ where f(x) being the polynomial divident, $g(x)$ being the divisor, $q(x)$ being the quotient and $r(x)$ is the remainder

When $g(x) = (x - c)$ then we obtain the relation;

$f(x) = (x - c)g(x) + r(x)$, clearly the maximum degree of $r(x)$ being $n - 1$, therfore $r(x)$ is a consta

$\implies \boxed{f(c) = r}$

The Factor Theorem (Corollary of Remainder Theorem)

Theorem:

$$\boxed{If\ f(c)\ is\ equal\ to\ zero\ then\ (x-c)\ is\ a\ factor\ of\ f(x),\ in\ other\ words\ c\ is\ a\ root\ of\ f(x)}$$

Proof:

From the relation $f(x) = (x - c)g(x) + r(x)$, when r(x) = 0 we obtain $f(x) = (x - c)g(x)$

Clearly, $(x - c)$ is a factor of the polynomial

Some Results:

$$\boxed{x^n - 1 = (x - 1)\left(\sum_{k=1}^{n} x^{n-k}\right)}$$

In the above formula, replacing x by x/y we obtain the relation, the number of terms in the second bracket being n

$$\boxed{x^n - y^n = (x-y)(x^{n-1} + x^{n-2}y + \cdots + xy^{n-2} + y^{n-1}) = \sum_{k=0}^{n-1} x^k y^{n-1-k}}$$

Using this we can obtain the common identities of difference of powers of 2,3,4,5 etc.

$$\boxed{x^3 - y^3 = (x-1)(x^2 + xy + y^2)}$$

$$\boxed{x^4 - y^4 = (x-1)(x^3 + x^2y + xy^2 + y^3)}$$

The expression for the sum of a geometric series can be derived from the first result aswell;

Let $a, ar, ar^2, ar^3, \ldots, ar^{n-1}$ be our given geoemetric progression.

$$S = a + ar + ar^2 + ar^3 + \cdots + ar^{n-1} = a\sum_{k=0}^{n} r^{n-k}$$

$$\implies \boxed{S = a\left(\frac{r^n - 1}{r - 1}\right)}$$

Synthetic Division

Theorem:

Synthetic Division is a method to divide any polynomial by a linear factor $(x-c)$ to obtain the value of the polynomial at a value c, without computing the value of $f(c)$ which is tedious.

Let $f(x) = \sum_{k=0}^{n} a_k x^{n-k}$ being divided by $(x-c)$. Let the quotient $q(x) = \sum_{k=0}^{n-1} b_k x^{n-1-k}$
From the remainder theorem our polynomial is $(x-c)q(x) + f(c)$ (1)

Comparing the coefficients of (1) and $f(x)$ we obtain the following relations:

$b_0 = a_0,\ b_1 = a_1 + cb_0,\ b_2 = a_2 + cb_1,\ \ldots,\ b_{n-1} = a_{n-1} + cb_{n-2}$ and $r = a_n + b_{n-1} = f(c)$

Use of the method:

- We write the coefficients of all powers of x including the missing powers as-well
- Starting from the first coefficient, we bring it down the line, multiply it with the value at which the polynomial is being evaluated.
- The term is then written below the second coefficient, added and then multiplied with c again, and this process is repeated, until the last term, where we obtain the remainder of the polynomial. The coefficients in the row below are of the quotient polynomial.

e.g. Evaluate $P(x) = x^4 + 3x^3 - 2x - 5$ at $x = 2$.

Since $x^4 = x \cdot x^3 = 2x^3$, the first two terms combine to give $5x^3$. Multiplying $5x^3$ by 2 and combining with the next term $-2x$ gives further partial sums, until we reach the final value.

$$P(2) = 1 \cdot 2^4 + 3 \cdot 2^3 + 0 \cdot 2^2 - 2 \cdot 2 - 5 = 31$$

We arrange the coefficients (including a 0 for the missing x^2 term) and carry out the synthetic division against $c = 2$:

The bottom row (excluding the last entry) gives the coefficients of the quotient, and the last entry is the remainder $P(2)$:

$$P(x) = (x-2)\big(x^3 + 5x^2 + 10x + 18\big) + 31$$

Factored form of Polynomials

Considering the following polynomial

$$f(x) = c_0x^n + c_1x^{n-1} + \cdots + c_{n-1}x + c_n$$

If $f(x) = 0$ has a root α_1 then from the factor theorem it has a factor $(x - \alpha_1)$, therefore

$$\implies f(x) = (x - \alpha_1)Q(x)$$

Evidently the degree of $Q(x)$ would be $n - 1$.Now suppose $Q(x) = 0$ has a root α_2 then

$$\implies f(x) = (x - \alpha_1)(x - \alpha_2)Q_1(x)$$

And so on we obtain;

$$\implies \boxed{f(x) = c_0(x - \alpha_1)(x - \alpha_2)\ldots(x - \alpha_n)} \qquad (1)$$

Now this expression has been mechanically deduced from the factor theorem. Deducing some more results;

Suppose that $f(x)$ of degree n has n distinct roots $\alpha_1, \alpha_2, \alpha_3 \ldots \alpha_n$ then from the factor theorem If $Q(x) = 0$ has the root α_2 then

$$\implies f(\alpha_2) = (\alpha_2 - \alpha_1)Q(\alpha_2)$$

From where $Q(\alpha_2) = 0$ and similarly if α_3 is a root of $f(x) = 0$ then repeating the same process

$$\implies f(\alpha_3) = (\alpha_3 - \alpha_2)(\alpha_3 - \alpha_1)Q_1(\alpha_3)$$

From where it is easily deducible that $Q_1(\alpha_3) = 0$

And so on we obtain the original factored form in (1) and our assumptions have been justified

If $f(x) = 0$ had any one more root other than $\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_n$ say α_{n+1} then it from the factor theorem it would have another factor in its factor form, but since a it can only have n factors, therefore we prove the theorem that
Theorem:

$$\boxed{Any\ polynomial\ \mathbf{f(x)}\ in\ x\ of\ degree\ \mathbf{n}\ can\ not\ have\ more\ than\ \mathbf{n}\ roots\ for\ the\ equation\ \mathbf{f(x) = 0}}$$

Multiplicity

Any root can occur more than one time in the factored form of a polynomial, the number of occurrence of the root being called its multiplicity

$$f(x) = c_0(x-\alpha_1)^{m_1}(x-\alpha_2)^{m_2}(x-\alpha_3)^{m_3}\dots(x-\alpha_k)^{m_k}$$

Here α_1 is called a root of multiplicity m_1 of $f(x)=0$, and so on.

$$m_1 + m_2 + m_3 + \cdots + m_k = n$$

which is pretty obvious.
The multiplicity is counted when counting the number of roots.